

2022 Spring ORF307 1st Midterm Exam

Instructions

Date and time: March 03, 2021, between 1:30pm – 2:50pm

Exercises: there are 3 problems worth a total of 100 points.

Exam rules

- You are allowed a single sheet of paper, double-sided, hand-written or typed.
- You cannot communicate with anyone during the exam.
- All questions should be answered in the booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine.
- Please remember to write your name below. On the next page, please write out and sign the following pledge: *“I pledge my honor that I have not violated the honor code or the rules specified by the instructor during this examination”*.

Name

Honor pledge

Problem 1 (30 points)

In least squares, the objective (to be minimized) is

$$\|Ax - b\|^2 = \sum_{i=1}^m (a_i^T x - b_i)^2,$$

where a_i^T are the rows of A , and the n -vector x is to be chosen. In the *weighted constrained least-squares* problem, we solve the problem

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^m w_i (a_i^T x - b_i)^2 \\ \text{subject to} & Cx = d \end{array}$$

where w_i are given positive weights. The weights allow us to assign different weights to the different components of the residual vector.

1. (10 points) Show that the weighted least-squares objective can be expressed as

$$\|D(Ax - b)\|^2$$

for an appropriate diagonal matrix D . This allows us to solve the weighted constrained least-squares problem as a standard constrained least-squares problem, by minimizing $\|\tilde{A}x - \tilde{b}\|^2$ subject to the constraint $Cx = d$, where $\tilde{A} = DA$ and $\tilde{b} = Db$,

2. (10 points) Show that when A has linearly independent columns, so does the matrix $\tilde{A}$.
3. (8 points) Give the KKT linear system for the weighted constrained least squares problem, in terms of A , b , C , d , x^* , z^* , and the matrix W , where W is defined to be $\mathbf{diag}(w)$, the diagonal matrix with the weights as its diagonal.
4. (2 points) Is the KKT matrix invertible? Explain. You may assume that the matrix C is wide with linearly independent rows.

Problem 2 (35 points)

Consider a linear dynamical system with dynamics

$$x_{t+1} = Ax_t + Bu_t, \quad t = 1, 2, \dots$$

with state n -vector x_t and input m -vector u_t . Given an initial point x_1 and final time K , our goal is to find an input sequence $u_1, \dots, u_{K-1}$ that results in $x_K = 0$. Choosing an input sequence that takes the state to the desired operating point at time K is called *regulation*.

1. (10 points) Define the *controllability matrix* as:

$$C = \begin{bmatrix} A^{K-2}B & A^{K-3}B & \dots & AB & B \end{bmatrix},$$

where $A^i = AA \cdots A$ (multiplied i times) and define the $m(K-1)$ -vector

$$\tilde{u} = (u_1, u_2, \dots, u_{K-1}),$$

representing the input sequence. By iteratively applying the linear dynamics, express x_K in terms of $x_1, A, K, C, \tilde{u}$.

2. (10 points) When searching for an input sequence, we will search for what we call the *minimum energy regulator*. That is, our goal is to find a sequence of inputs that achieves the desired state x_K and minimizes

$$\|u_1\|_2^2 + \dots + \|u_{K-1}\|_2^2.$$

Set the final state constraint to $x_K = 0$ using the formula derived in Part 1. Write the problem of minimizing $\|u_1\|_2^2 + \dots + \|u_{K-1}\|_2^2$ subject to this constraint as a constrained least squares problem. Your problem should be in terms of $x_1, A, K, C, \tilde{u}$.

3. (5 points) Write the KKT linear system for the constrained least squares problem from Part 2.
4. (10 points) Solve the KKT system to derive an explicit formula for the optimal $\tilde{u}$ in terms of x_1, A, K, C . You may assume that CC^T is invertible.

Problem 3 (35 points)

Consider the following linear program.

$$\begin{array}{ll}\text{minimize} & 2\|x\|_\infty \\ \text{subject to} & \mathbf{1}^T x \geq 1 \\ & x \geq 0\end{array}$$

1. (10 points) Write the LP in inequality form. You do not have to write it the most compact form.
2. (5 points) Draw the feasible region in 2D in terms of x_1 and x_2 (and not any additional variables you may have added in part 1).
3. (5 points) Draw contour lines for the objective equal to 1, 2, and 4.
4. (5 points) Identify the optimal solution x^* and justify it graphically.
5. (10 points) Write the LP in standard form. You do not have to write it in the most compact form.

